

Customer Insights and Prediction System – Enhanced Project Document

Project Overview

The Customer Insights and Prediction System is a full end-to-end data solution designed to analyze customer behavior, produce actionable insights, and build predictive machine learning models. The enhanced version below restructures the proposal with improved formatting, tables, and clarity.

Team Members & Roles

Team Member	Role
Youssef Ali Hegazy	Team Leader – Machine Learning
Osama Essam Azab	Data Preprocessing
Mohammed Hani Ebraheem	Data Collection & Power BI Dashboard
Ahmed Tarek Elmenoufy	Database
Tomas Amir Gerges	Airflow & Automation
Mostafa Sobhy Mahmoud Shehab	Data Warehouse

Objectives

- Collect raw customer data from multiple sources.
- Perform data cleaning and preprocessing for high-quality datasets.
- Store structured data in a well-designed data warehouse.
- Implement SQL-based data analysis and reporting.
- Develop automated workflows using Apache Airflow.
- Build ML models for prediction and behavioral analysis.
- Visualize business KPIs through an interactive dashboard.

Tools & Technologies

- Python, Pandas, NumPy
- SQL (MySQL / PostgreSQL)
- Data Warehouse Design
- Apache Airflow

- Power BI Visualization
- Machine Learning (Scikit-Learn)
- GitHub for version control

Milestones & Completion Status

Milestone	Status
Data Collection	Completed
Preprocessing & Cleaning	Completed
Data Warehouse Setup	Completed
Airflow Pipelines	Completed
ML Model Training	Completed
Dashboard & Reporting	Completed

Key Performance Indicators (KPIs)

Data Preprocessing

- 100% missing and duplicate data handled
- Efficient script runtime within expected thresholds

SQL Integration

- ≥95% query accuracy with correct results
- Fast query execution within defined limits

Visualization

- Dashboard load time < 3 seconds
- ≥90% KPIs and metrics visually represented

Presentation

- 100% completeness of all required documents
- Stakeholder clarity/feedback score ≥ 4/5